

GENERAL SPECIFICATION G4.56

Tank Mixing Systems

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G4.56 TANK MIXING SYSTEMS

G4.56.1 General

- (a) Definitions - Terminology pertaining to pumping unit performance and construction shall conform to the ratings and nomenclature of the Hydraulic Institute Standards.
- (b) Submittals
 - (i) Mixing system schematic illustrating especially any proprietary design of the mixing system process and piping configuration;
 - (ii) Shop Drawings;
 - (iii) Make, model, weight, and power kilowatt rating of each equipment assembly;
 - (iv) Complete mixer and motor catalogue information, descriptive literature, specifications, and identification of materials of construction subject to S.O.'s approval;
 - (v) Performance data curves showing head, capacity, power demand, and mixer efficiency over the entire operating range of the mixer, from shutoff to maximum capacity. Indicate separately the head, capacity, horsepower demand, overall efficiency, and minimum submergence required at the guarantee point;
 - (vi) Computational fluid dynamics (CFD) analysis that models the flows within each tank;
 - (vii) Detailed structural, mechanical, and electrical drawings showing the equipment dimensions, size, and locations of connections and weights of associated mixing system and recirculation equipment; and
 - (viii) Standard Operational Procedures (SOP) including procedures for cleaning, flushing and maintenance.
- (c) The Contractor shall ensure that the selected manufacturer of the mixing system possesses past project references of at least 5 installations which have already been operated at optimum performance for 5 years or longer.

G4.56.2 Hyperboloid

- (a) The mixer shall be vertical mounted. Where applicable, design and construction of support, drive, shaft and bearing and gear box shall comply with the requirements of mechanical mixer as stated above.
- (b) The mixer shaft and impeller assemblies may be constructed from fiberglass reinforced plastic.
- (c) The mixer shall be provided with a single impeller of semi open hyperboloid design. Impellers shall be of a design, arrangement, and diameter and shall operate at a speed that will provide efficient and proper mixing of the wastewater at the design flow of the process treatment unit as specified.
- (d) The rotational speed shall not exceed 65% of the critical speed of the shaft (including impellers and appurtenances). Bottom steady bearings will not be acceptable.

- (e) Impeller shall be provided with removable arms to permit the impellers to be removed through the access hatches in the cover slab.
- (f) The impeller shall have hyperboloid shaped to ensure that oxygen concentration is uniformly distribute throughout the tank with minimum energy consumption. The mixing flow pattern shall ensure there is no sedimentation at the bottom of the tank.

G4.56.3 Submersible Propeller Type

- (a) The mixing unit shall be closed couple direct driven or integral gear driven submersible type. All components of the mixer, including motor and gear box shall be designed to suit the operation environment and condition. All components of the mixer including the motor and gear box shall be capable of continuous operation in air, even when completely unsubmerged.
- (b) A sliding guide brackets shall be provided. The mixer shall be guided by a single bracket which shall be capable of handling all thrust created by the mixer.
- (c) The proper blades shall be dynamically balanced and of non-clogging backward curve, the number of blades or the propeller shall be proposed and submitted for S.O.'s approval.
- (d) The seal shall be mechanical seal and maintenance free and easy to check and replace. Shaft seals lacking positively driven rotating members or conventional double mechanical sealed which use a common single or double spring and required a pressure differential to offer external pressure and effect sealing will not be acceptable.
- (e) All mating surfaces shall be machined and fitted with nitrile rubber O ring where watertight sealing is necessary.
- (f) The upper mast bracket shall be constructed with a position locking plate in conjunction with a lock pin to positively lock the mast in place at various operating angles and allow 180 degrees movement of the hoist and mast about a pivot point.
- (g) The mast shall be constructed structural tubing and the size of the tubing shall be recommended by the manufacturer.
- (h) The lower mast bracket shall support the entire weight of the mixer mast, mixer and hoist assembly, shall connect the mixer mast to the floor and shall include a mast pivot pin.
- (i) The mast assembly shall include a stainless steel 304 lifting cable, of same size specified for the hoist cable, which will connect to the hoist lifting cable using the quick disconnect latching mechanism.
- (j) The power cord shall be tethered to the lifting cable using stainless steel hardware. The tether hardware shall include heavy duty rubber hose to protect the power cord and siding carabiner that allows power cord to slide freely on the lifting cable.
- (k) The lifting cable shall be attached to the mixer with a lifting bracket or bridle chain which will allow the lifting cable to be centred over the mixer. A stainless steel bridle chain shackle shall be provided to connect the lifting cable to the lifting bracket or bridle chain.

- (l) A retaining chain shall extend from the top of the mixer mast to the mixer and shall allow the mixer to be located at any intermediate point on the mast. The bridle chain shackle shall connect the retaining chain to the mixer. All chain and shackle materials of construction shall be stainless steel 304.

G4.56.4 Pumped Jet (Vortex Jet)

- (a) The Contractor shall provide one complete mixing system for each tank equipped with mixing nozzles, piping, valves, mixing pumps, electric motors, supports, brackets and all associated ancillary equipment.
- (b) Means of redundancy such as additional mixing units of the same model or alternative means of mixing subject to the S.O.'s approval shall be provided such that it is possible to service a mixing system without shutting down the mixing tank.
- (c) The number, size, and locations of nozzles shall be designed to produce a rotational toroidal mixing pattern within the tank. Solids shall be effectively drafted by the nozzles' discharge to the outer 30 % of the tank into a peripheral helical rotation pattern following a toroidal contour with flow rising along the tank wall and descending into the middle of the tank. This shall sweep solids from the centre of the tank and prevent mounding at the tank centre.
- (d) Zone mixing that is either horizontally or vertically only shall not be acceptable but the mixing patterns shall create a homogeneous state throughout the entire volume suspending both organic and inorganic solids.
- (e) The manufacturer shall be responsible for determining the exact types of mixing pump and motor, mixing assembly quantity, location, and appropriate nozzle angles.
- (f) The manufacturer shall be responsible for confirming the specified pump capacity is adequate to provide the specified performance and provide the necessary pump equipment.
- (g) Tank configuration, dimensions and side water depth shall be as shown in the 3D Model.
- (h) Performance Requirements
 - (i) The tank mixing system shall positively and continuously mix the contents of the tanks. The variation in total solids concentration (TSS) throughout the tank shall not vary more than $\pm 10\%$ from the mean total solids for the tank when within the range of bulk concentrations specified;
 - (ii) Where required, the fluid temperature shall not vary more than $\pm 1\text{ }^{\circ}\text{C}$ from the mean average tank temperature;
 - (iii) The calculated active volume shall not be less than 90 % of the total tank volume; and
 - (iv) The Contractor shall submit the manufacturer's standard warranties and warranties for achieving the performance parameters specific to the intended application(s).

- (i) Nozzle Construction
 - (i) Nozzles shall be designed to be of an eccentric configuration and to discharge according to the manufacturer's recommended angle (eg. 5 deg., etc.) where applicable to enhance the sweeping pattern across the tank floor;
 - (ii) Nozzle orifice type and size shall be selected to provide optimum mixing system performance;
 - (iii) Nozzles shall have Van-Stone or slip style type flanged connection to allow 360 degrees adjustment;
 - (iv) Flow control vanes or straight taper length having a minimum length of ten pipe diameters shall be incorporated with the nozzles to improve discharge pattern where applicable;
 - (v) Where applicable, cathodic protection in the form of zinc anodes shall be attached to each base and wall bracket of the nozzle and piping or according to the manufacturer's recommendation;
 - (vi) Materials of construction for the nozzles shall be as follows:
 - 1) ASTM A536 glass-lined cast ductile iron hardened to minimum Rockwell C 73, with 26 mm minimum wall thickness with additional manufacturer's proposed coating as necessary; or
 - 2) ASTM A532 class II, Type C Chrome iron with a minimum Brinell hardness of 450/550 throughout the entire nozzle wall with additional manufacturer's proposed coating as necessary.

Notwithstanding the above materials of construction, the Contractor shall ensure that the manufacturer guarantees the corrosion resistance of the selected material for the intended specific application(s).

Certification shall be provided for the glass-lining quality to ensure pin-hole free and no discontinuity on the glass lining surfaces. Nozzles having any discontinuities or pin holes shall be rejected and replaced at no cost to the Board.
- (j) Tank Mixing Pump performance shall yield the output as recommended by the selected mixing system manufacturer to meet performance guarantees. The Contractor shall be under contractual obligation to ensure that this requirement is complied with.
- (k) On sludge holding tanks and other tanks, where specified, scum nozzle(s) shall be provided to break up and re-entrain scum and other floating material into the bulk liquid.
- (l) Submersible mixing pumps shall be used where these are mentioned in the particular specifications for specific applications, or as shown in the P & IDs or in the 3D model. Requirements for such pumps are as follows:
 - (i) Centrifugal non-clog type;
 - (ii) Chopper impeller;
 - (iii) Submersible electric motor with tandem mechanical seals in oil bath, dual moisture sensing probes and two normally closed automatic resetting thermostats connected in series and embedded in adjoining phases;

- (iv) Suitable for continuous operation at full load while motor is completely submerged, partially submerged or totally non-submerged. Use of shower systems, secondary pumps or cooling fans to cool the motor shall not be acceptable; and
- (v) Guide rail system consisting of two steel guide rails, quick discharge connector, support duckfoot or equivalent all of corrosion resistant materials of construction.
- (m) All materials of construction of mixing pumps shall be guaranteed for corrosion resistance with respect to the pumped fluid and the environment of installation whether dry or submerged. A comprehensive listing of the pump components with the respective materials of construction shall be submitted to the S.O. for approval.
- (n) Relevant sections of the General Specification – Pumps shall be read in conjunction with this specification.
- (o) Piping
 - (i) The piping system for the vortex jet mixing system shall be complete in all aspects with all necessary fittings, valves, adaptors, supports and brackets;
 - (ii) Piping and fasteners material of construction shall be stainless steel SS316L as a minimum;
 - (iii) The pressure rating of the piping shall be adequate to withstand at least 1.5 times the maximum operating pressure or the jet pump shutoff head whichever is higher; and
 - (iv) Stainless steel SS316L pipe wall thickness schedule shall be at least SCH40S for piping with threaded connections.

G4.56.5 Air Mixing

- (a) Air mixing shall be used to achieve homogeneous mixing of the contents of a tank with or without the addition of chemicals and to keep the contents in suspension to avoid settling.
- (b) Air mixing technology as employed in mixing the contents of sludge digester holding tanks shall employ externally compressed air to be delivered and released into the contents through a series of mixing points within the tank at timed sequences and durations.
- (c) The mixing flow pattern in the tank shall be stable, uniform and shall rotate the tank contents.
- (d) The mixing flow pattern shall consist of the flow of air bubbles which are released from an outer ring of mixing points flowing in one direction and those released from an inner ring flowing in the reverse direction.
- (e) Each mixing point shall release air over a duration of approximately 3 min. or as recommended by the manufacturer/supplier subject to approval by the S.O..
- (f) The mixing shall be compatible with odorous air extraction from the tank.
- (g) The mixing shall not employ moving/wear parts in contact with the tank contents being mixed.

- (h) The design consideration and performance parameters for air mixing shall be as follows:
 - (i) Volume of tank contents that is actively mixed: Minimum 90 %
 - (ii) Minimum mixing depth: 4 m
 - (iii) Maximum Dry solids content of sludge mixable: 8 % D.S.
 - (iv) Minimum energy input to mixed contents: 2 W/m³
- (i) The arrangement layout drawings and mounting footprint drawings of the compressor system with the associated acoustic enclosure showing suction intake, delivery outlet, ventilation air intake or/and exhaust, local control panel, instruments, valves, piping, supports and all appurtenances shall be submitted to the S.O. for approval.
- (j) Complete equipment system datasheets, technical brochures, control logic, wiring diagrams, erection/installation, Operation and Maintenance manuals and instruments settings shall be submitted to the S.O..
- (k) The arrangement of the air mixing points at the hoses' ends and the layout of the hoses within each tank shall be submitted to the S.O. for approval.
- (l) Fully detailed dimensioned general arrangement drawings shall be submitted to the S.O. for all pipe and valve supports which shall include but not limited to the following:
 - (i) Adjustable pipe supports able to accommodate ± 30 mm variation in ∇ Bottom of Pipe (/BOP) elevation and ± 30 mm variation in longitudinal and transverse shift from axis of pipe; and
 - (ii) Pipe supports configured of uni-struts, flat plats, angle-bars, universal H-beams, universal C-channels, pipe stanchions, pipe clips, pipe clamps, half clamp blocks, hexagon rail nuts, U-bolts, mounting rails, etc.
- (m) The Contractor shall be responsible for all coordination between tank and air mixing system manufacturer/supplier to ensure that all installation work sequences and interfaces including pipe support attachments and wall surfaces are compatible.
- (n) The Contractor shall be responsible to ensure that tank design information and data relevant to the air mixing system manufacturer/supplier and that air mixing information and data relevant to the tank manufacturer/supplier are communicated between the two in timely fashion.
- (o) Chemical anchors with stainless steel SS316 studs shall be used for fixing pipe supports and other appurtenances within the tank. The epoxy adhesive shall be of the type that give the adequate bond strength and that cures to become a hardened compound impervious to liquid or gas and unaffected by the contents of the tank.
- (p) All other metallic pipe supports including uni-struts, support fasteners and accessories shall be made of stainless steel SS316.
- (q) All pipe support bolted fixings within tanks shall be with double nuts or fitted with split-ring spring washers.

- (r) The key features for the compressor system to delivering the mixing air are as follows:
- (i) Lubrication/Cooling method: Air-cooled oil-free.
 - (ii) Free Air Delivery (FAD): (Refer to Particular Specifications for each specific application.)
 - (iii) Discharge flow: (Refer to Particular Specifications for each specific application.)
 - (iv) Discharge pressure: (Refer to Particular Specifications for each specific application.)
 - (v) Acoustic enclosure: To be provided with guarantee of noise level not exceeding 85 dB(A) at 1 m away from enclosure.
- (s) As a minimum, the following appurtenances shall be provided:

No.	Item	Qty.	Remark
1.	Pressure Relief Valve	1 no. per compressor	Size and type to be proposed and submitted for approval by S.O..
2.	Non-Return Valve	1 no. per compressor	
3.	Delivery Pipework Drain Ball Valve	1 no. per compressor	Size to be proposed and submitted for approval by S.O..
4.	Delivery Pipework Isolation Ball Valve	1 no. per compressor	
5.	Solenoid valve	(Total quantity as required)	
6.	Acoustic Enclosure Temperature Switch	1 no. per compressor acoustic enclosure	Instrument type and settings to be proposed and submitted for approval by S.O..
7.	Delivery Temperature Switch	1 no. per compressor	
8.	Suction Pressure Switch	1 no. per compressor	
9.	Delivery Pressure Switch	1 no. per compressor	
10.	Suction silencer	1 no. per compressor	Optional depending on noise level.
11.	Discharge silencer	1 no. per compressor	

No.	Item	Qty.	Remark
12.	Suction Air Filter	1 no. per compressor	Filtered particle size as dictated by air quality of the mixing air.
13.	Discharge air Filter	1 no. per compressor	
14.	Flexible bellow	1 no. per compressor	

- (t) Material of construction for the mixing air pipework shall be as follows:

No.	Pipe/Hose Identity	Material of Construction
1.	Compressor discharge piping up to air mixing manifold	SCH. 10S SS316 for pipework having non-threaded connections
		SCH. 40S SS316 for pipework having threaded connections
2.	Mixing air hose from air mixing manifold to mixing point within tank	Flexible EPDM hose

- (u) The General Specifications G4.2 Pipework and G4.5 Compressed Air Systems shall be referred to for additional information as necessary.

END OF SECTION